



Advanced Web Programming

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Lab 4

Assignment 1

Enhance the to-do list app from Lab 3 by adding “Assign to member”, “Due date” fields.

My To Do List			
Title...	Member	03/16/2025	
Task 1 - Giàu - 15.1.2025			x
✓ Task 2 - Thanh - 16.1.2025			x
Task 3 - Nhi - 16.2.2025			x

Assignment 2

Use **useReducer** to build a shopping cart management.

Products

Laptop - \$1000

Add to Cart

Phone - \$500

Add to Cart

Headphones - \$100

Add to Cart

Shopping Cart

Headphones - \$100 x 1

Remove

+

-

Phone - \$500 x 2

Remove

+

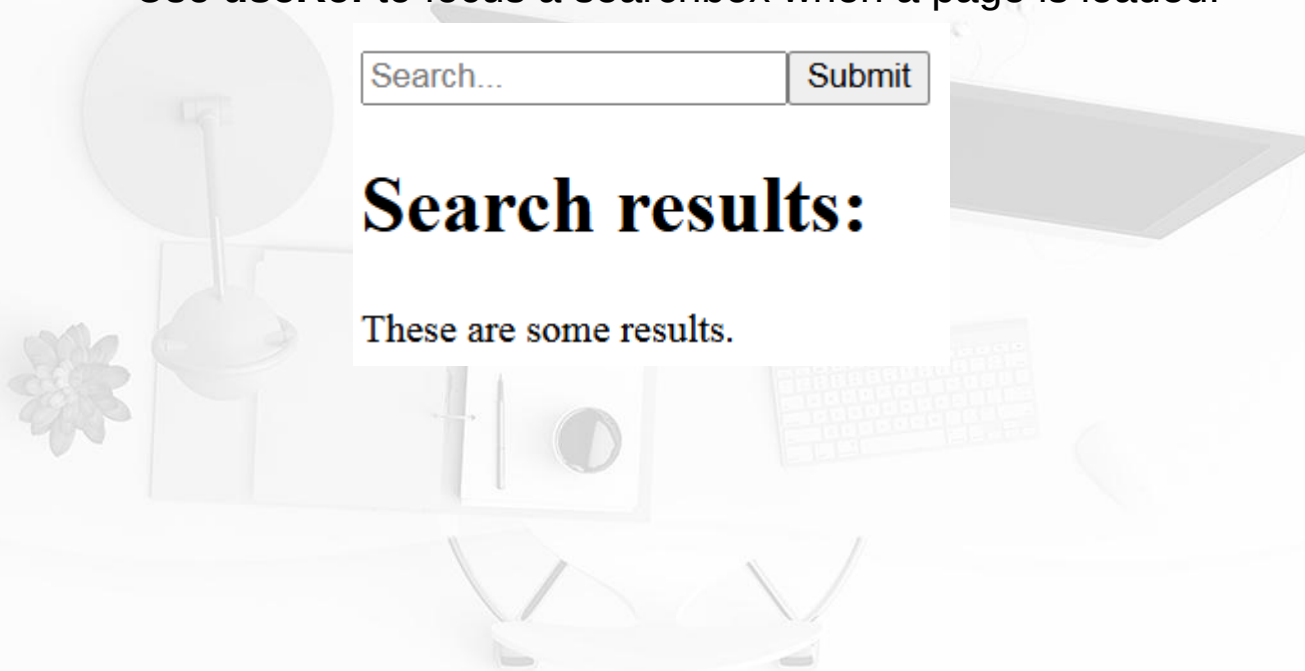
-

Total: \$1100

Clear Cart

Assignment 3

Use **useRef** to focus a searchbox when a page is loaded.



Search... Submit

Search results:

These are some results.

Assignment 4

A web app renders a large dataset (e.g., 10,000+ users) and has a user filtering feature.

Use **useMemo** to optimize the **filterUser()** function, ensuring that the component does not re-render or refetch data when the state of an **additional option** changes.

Search users... ☐ Additional Option

- User 1
- User 2
- User 3
- User 4
- User 5
- User 6
- User 7



Elements

Console

Sources



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Filter

[Download the React DevTools for a better experience.](#)

2 Fetched

Assignment 4

- This is sample code to generate the user list.

```
const users = Array.from({ length: 10000 }, (_, i) => ({  
  id: i + 1,  
  name: `User ${i + 1}`,  
}));
```

- This is sample code to change state of the Additional Option.

```
{/* const [check, setCheck] = useState(false); */}  
<input type="checkbox" checked={check}  
onChange={(e) => setCheck(e.target.checked)} /> Additional Option
```